

1. For a vector $\mathbf{x} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$ and the subspace U spanned by the basis vectors $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $\mathbf{b}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$, which of the following statements are true?

4 points

You can use the formula slide that comes with the corresponding lecture.

- ☐ The projection of $\mathbf{x}$ onto U is $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.
- ☐ The coordinates of the projected point with respect to $\mathbf{b}_1, \mathbf{b}_2$ are $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$.
- ☒ The projection matrix is $\frac{1}{6} \begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}$.
- ☒ The coordinates of the projected point with respect to $\mathbf{b}_1, \mathbf{b}_2$ are $\begin{bmatrix} 5 \\ -3 \end{bmatrix}$.
- ☐ The projection matrix is $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 4 \end{bmatrix}$.
- ☐ The rank of the projection matrix is 1.
- ☒ The projection matrix is symmetric.
- ☒ The projection of $\mathbf{x}$ onto U is $\begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$.
- ☐ The projection matrix is not symmetric.

2. Project $\begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}$ onto the subspace spanned by $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

1 point

You can use the formula slide that comes with the corresponding lecture.

- ☒ $\begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}$
- ☐ $\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$
- ☐ $\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$
- ☐ $\begin{bmatrix} 6 \\ 4 \\ 4 \end{bmatrix}$

3. 1. Project $\begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}$ onto the subspace U_1 spanned by $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$.

1 point

2. Project the result from 1. onto the subspace spanned by $\begin{bmatrix} -10\sqrt{6} \\ -4\sqrt{6} \\ 2\sqrt{6} \end{bmatrix}$. What is the final projection?

Hint: For step 2, you do not necessarily need to compute anything.

You can use the formula slide that comes with the corresponding lecture.

- ☐ $\begin{bmatrix} 5 \\ 2\sqrt{6} \\ -1\sqrt{6} \end{bmatrix}$
- ☒ $\begin{bmatrix} 10 \\ 4 \\ -2 \end{bmatrix}$
- ☐ $\begin{bmatrix} 5 \\ 2\sqrt{6} + 1 \\ -\sqrt{6} + 2 \end{bmatrix}$